

Optimole Repository Layout

Optimole is organized around the build flow for generated game projects:

```
```text
User
-> NestJS API Gateway
-> LLM-generated Game Specification JSON
-> Save spec
-> Publish build job to RabbitMQ
-> Optimole Build Worker
-> Validate spec, generate project, invoke Unreal/Unity tools
-> Upload artifacts to S3/MinIO
-> API updates job status
-> Client downloads the build from CDN/Object Storage
```
```

Canonical Folders

```
```text
api/ NestJS API Gateway
frontend/ User-facing build console
worker/ Build worker runtime and engine service adapters
browser-engine/ Full browser playable runtime served by the API
shared/ DTOs, types, and interfaces shared across services
spec-engine/ JSON schemas for generated game specifications
templates/ Browser, Unreal, and Unity project templates
docker/ Local infrastructure compose files
scripts/ Repository maintenance and workflow scripts
```
```

Infrastructure Shape

```
```text
Internet
-> Cloudflare / Load Balancer
-> Optimole API Gateway (NestJS)
-> AI Generation Cluster + Build Queue Cluster
-> Redis / RabbitMQ
-> Unreal / Unity Workers
-> Object Storage (S3/MinIO)
-> CDN Download Server
```
```

Local Service Commands

For the full local beta stack:

```
```bash
scripts/start-optimole.sh --install
```
```

After dependencies are installed, use:

```
```bash
scripts/start-optomole.sh
```
```

This starts Docker infrastructure, the API gateway, build worker, and frontend.
Open `http://localhost:3000` to use Optomole.

```
```bash
cd frontend
npm install
npm run dev
```
```

```
```bash
cd api
npm install
npm run start:dev
```
```

```
```bash
cd worker
npm install
npm start
```
```

```
```bash
docker compose -f docker/compose.dev.yml up --build
```
```

Developer Rule Of Thumb

Use:

`templates/`

for compiler-facing metadata.

Use:

`browser-engine/templates/`

for actual gameplay code.

A complete new browser template needs both.

Deprecated Material

Older frontend/runtime experiments, generated build outputs, logs, cache folders, legacy API code, and archived docs were moved into `OptoDeprecated/`. Nothing was deleted during the cleanup.